ISA Data Fluency

IMBA Session 1

"In order to scale with AI, you first have to become data-driven." –Richard Benjamins

Professor Christina Stathopoulos, September 2026

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Course Introduction

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Christina Stathopoulos

Founder, Data & AI Evangelist
Dare to Data

Ex-Google, 5+ years…
- Analytical Lead, **Waze @Google NYC**
- Data Specialist, **Google Spain**

Adjunct Professor of Data & AI
IE Business School | IE University

Show Host **@O'Reilly Media**
Podcast Host **@EM360Tech**
Instructor **@LinkedIn Learning**

Master in Business Analytics & Big Data
IE School of Science & Technology

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CONTACT ME

Preferred mode of contact during course is via Blackboard

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IMPACT SKILLS ACCELERATOR

Impact Skills Workshops for catalyzing action and results in any workplace

IE Business School Framework - Thinking Skills + Behavioral Skills + Digital Skills – our terminology, but consistent with what the world of work demands



THINKING

Formulate, analyze and generate solutions to challenges & plan and carry out work

Critical Thinking

Address complexity, make decisions and sound judgements with an open mindset

Creative Thinking

Generate solutions through creative strategies and a user-centric approach

Problem Solving

Tackle difficult challenges through analytical strategies

Project Management

Design, coordinate, and implement complex projects



BEHAVIORAL

Sharpen your abilities to overcome challenges and move ideas forward

Self Transformation

Understand yourself (self awareness) and be open to get transformed by gaining behavioral agility (self management)

High-performance Teambuilding I&II

Build and maintain diverse, efficient & self healing teams

Storytelling & Public Speaking

Create stories for leveraging mind-chanching communication and effectively present ideas in public

Power & Influence

Manage negotiation to build sustainable alliances, solve conflicts and address power games



DIGITAL

Run business based on digital insights and data

Data Fluency

Gather, organize, analyze, and visualize data to make your case

AI for Productivity

Learn how to use Generative AI tools to increase your competencies and booster your productivity

Data Fluency - Objective

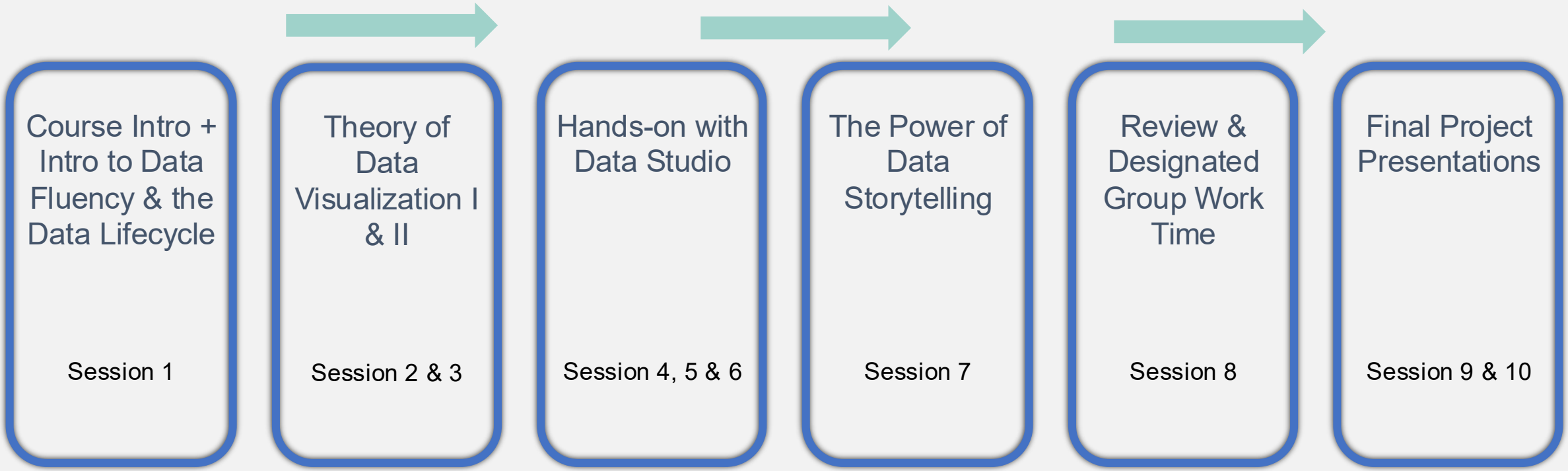


Equip students with the knowledge and skills to effectively navigate a business world increasingly driven by data.

Particular focus on ‘the last mile’ of data insights delivery through data visualization and data storytelling.

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Course Breakdown



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Grading Breakdown

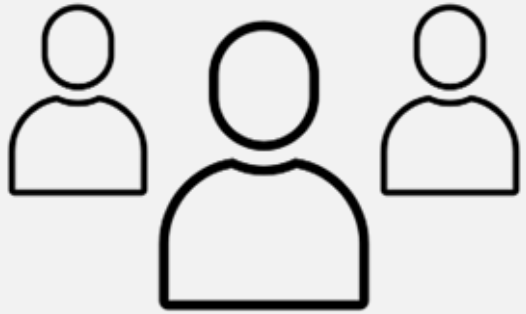
Graded
***included in GPA**
but no enforced curve!



Individual – 50%

Attendance + Active Participation – 20%
(8+ of 10 sessions required)

Midterm Exam – 30%



Group – 50%

Final Group Presentation

Note that I will take attendance at the beginning of each session.
If you must miss class for an excused (IE-approved) reason, arrive late or need to leave early, it is your responsibility to write Program Management to get attendance credit.

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Grading

IMPORTANT

All assignments are graded using the same criteria:

- **Honors** (91 - 100): Exceptional mastery, originality & insightful application of course concepts
- **Excellence** (81 - 90): Strong understanding & application of concepts with minor errors
- **Proficiency** (71 - 80): Meets requirements through adequate understanding of course concepts with minor errors
- **Pass** (61 - 70): Sufficient understanding & application of concepts, but lacks depth & contains significant errors
- **Fail** (0 - 60): Fails to meet basic requirements with minimal understanding and major mistakes

2 dates to remember:

September 17th	**J-ISA**	October 28th
midterm exam		group presentation

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My Approach



Lecturing to teach core concepts

Using examples to connect theory to practice

Class discussion, real-world examples, videos, hands-on practice

And… try to have some fun ☺

Please review the (Gen)AI policy outlined in the syllabus*

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Class Logistics



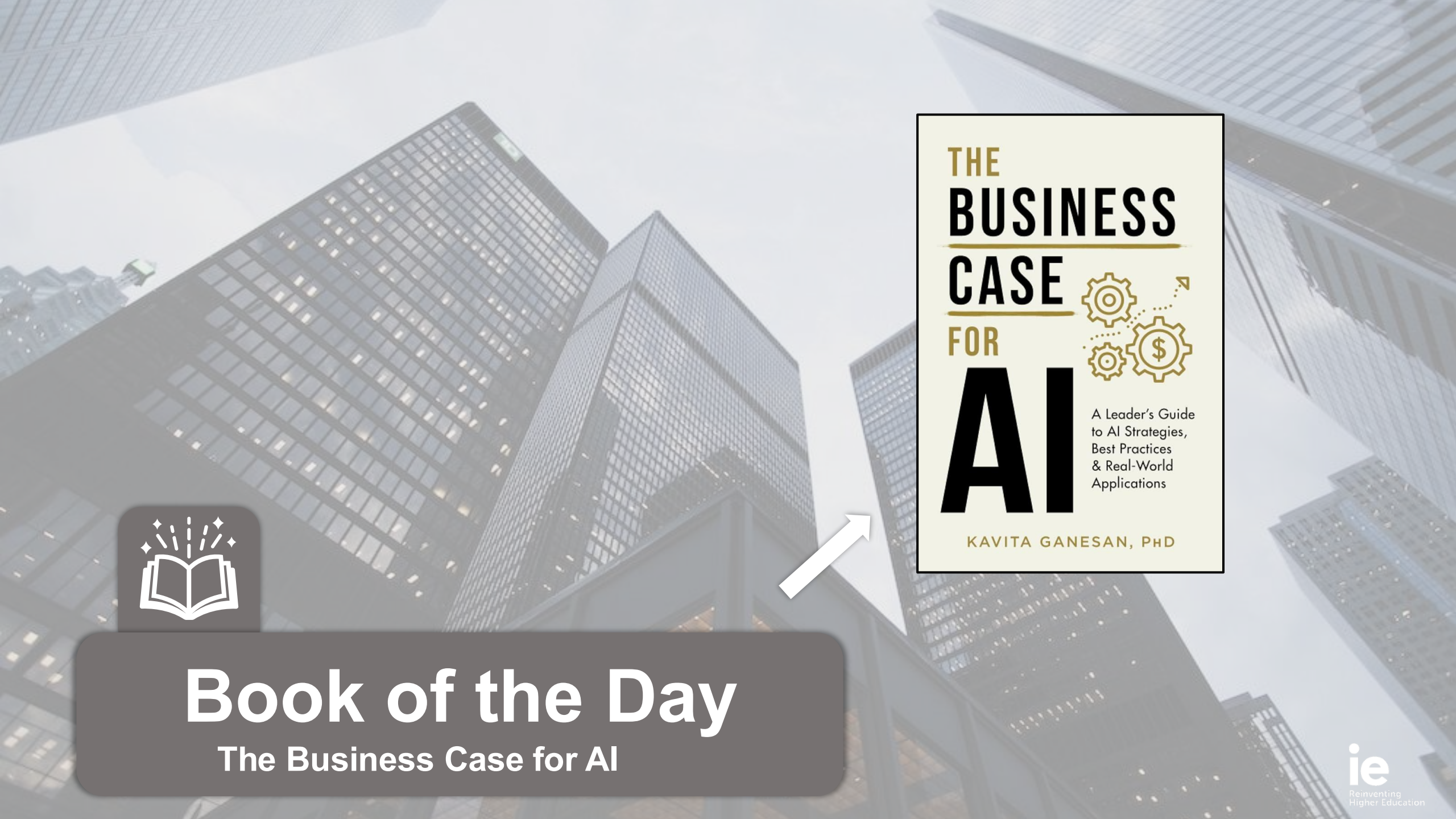
Fully Here – Be present and pay attention.

Try – We will debate, do exercises. Don't be afraid to try new things and fail.

Curious – Be curious about our course content, don't be afraid to ask questions. You can raise your hand at anytime.

On Time – We start all classes on time so be punctual.

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Book of the Day

The Business Case for AI

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To truly scale the value of AI...

We must invest in how we process & share our data.

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Session 1 Agenda



Data Fluency



The Data Lifecycle

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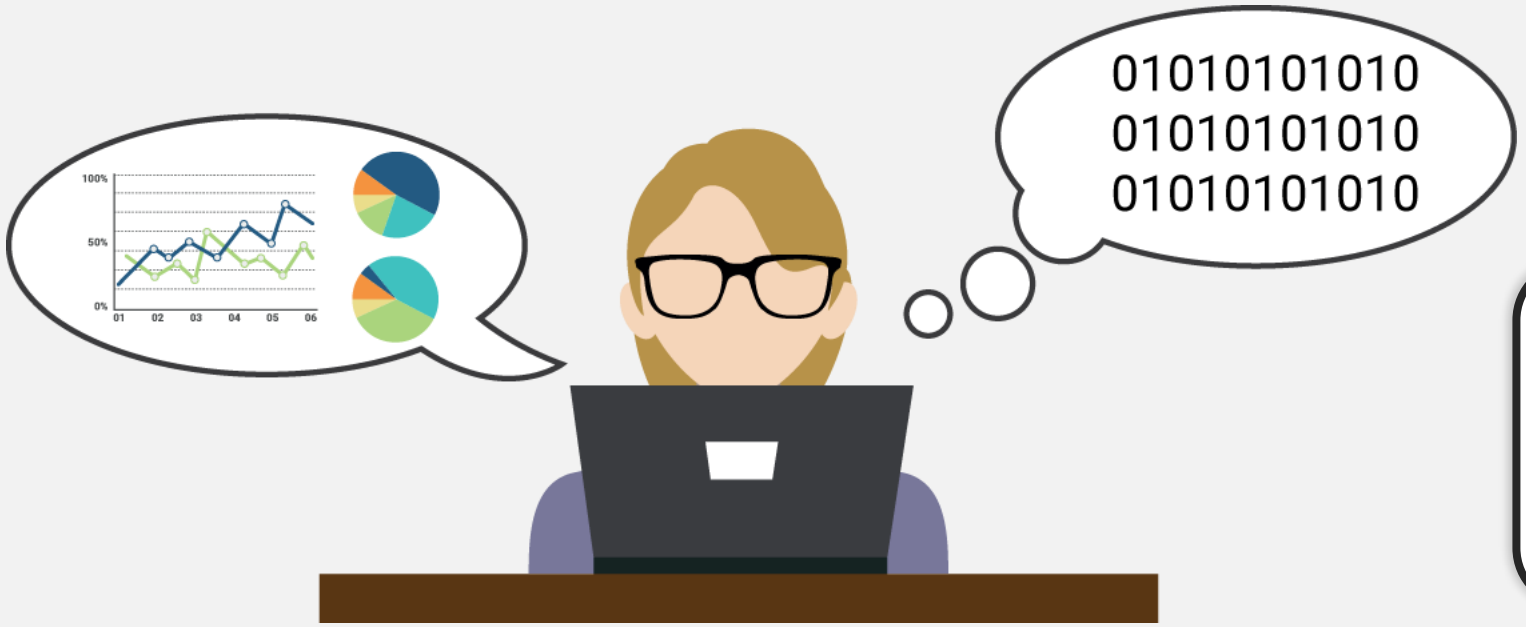


Data Fluency

Understanding, Interpreting & Using Data

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What is Data Fluency?



This is *necessary* in order to…

- interpret or use data results for meaningful business applications
- establish a data-driven culture within your organization

The ability (and confidence!) to understand, interpret, and use data proficiently.

In today's business landscape, everyone should possess this.

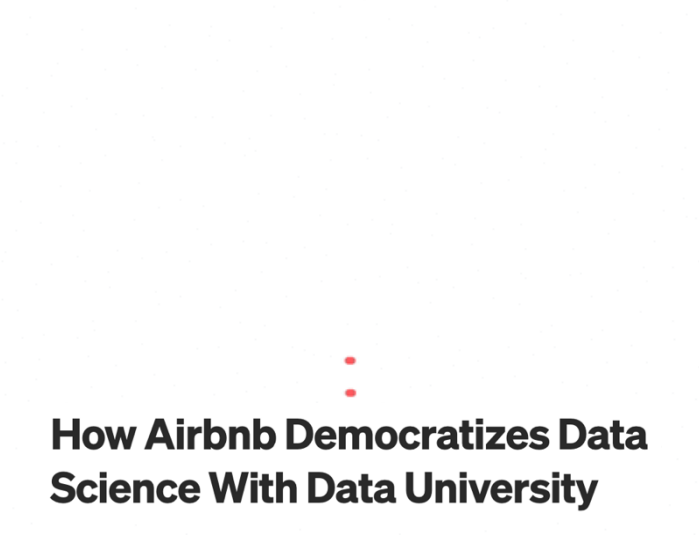
How Data Fluent are YOU?

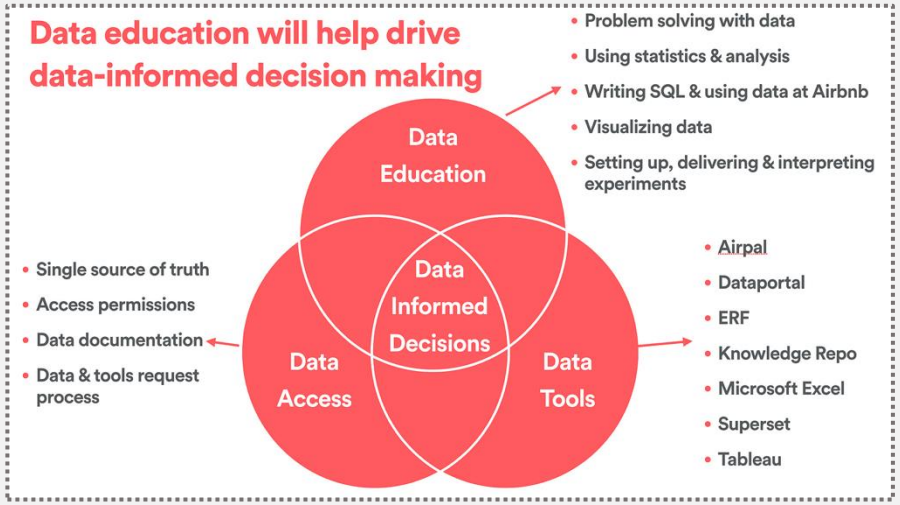
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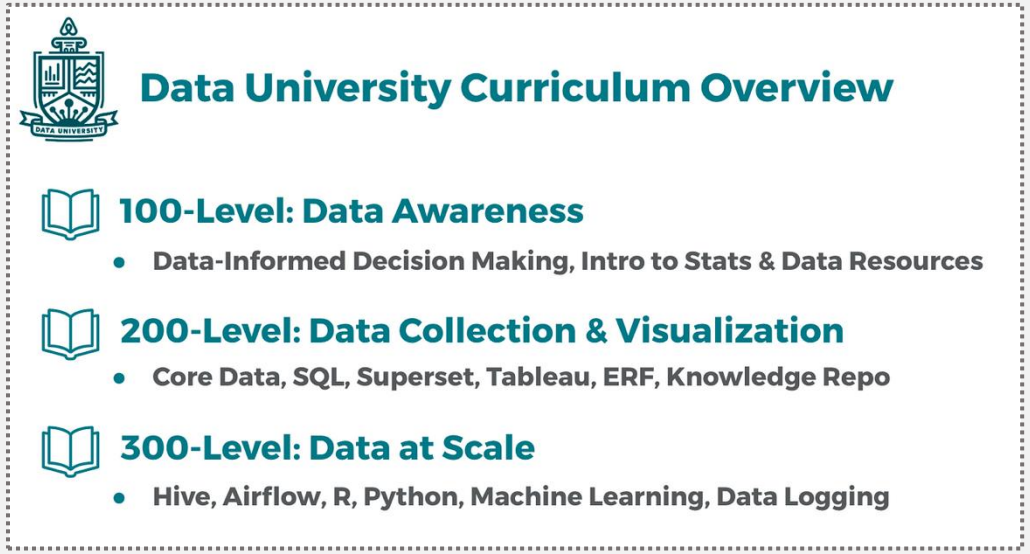
Data skills should not be limited to the data teams…

Data Fluency for all









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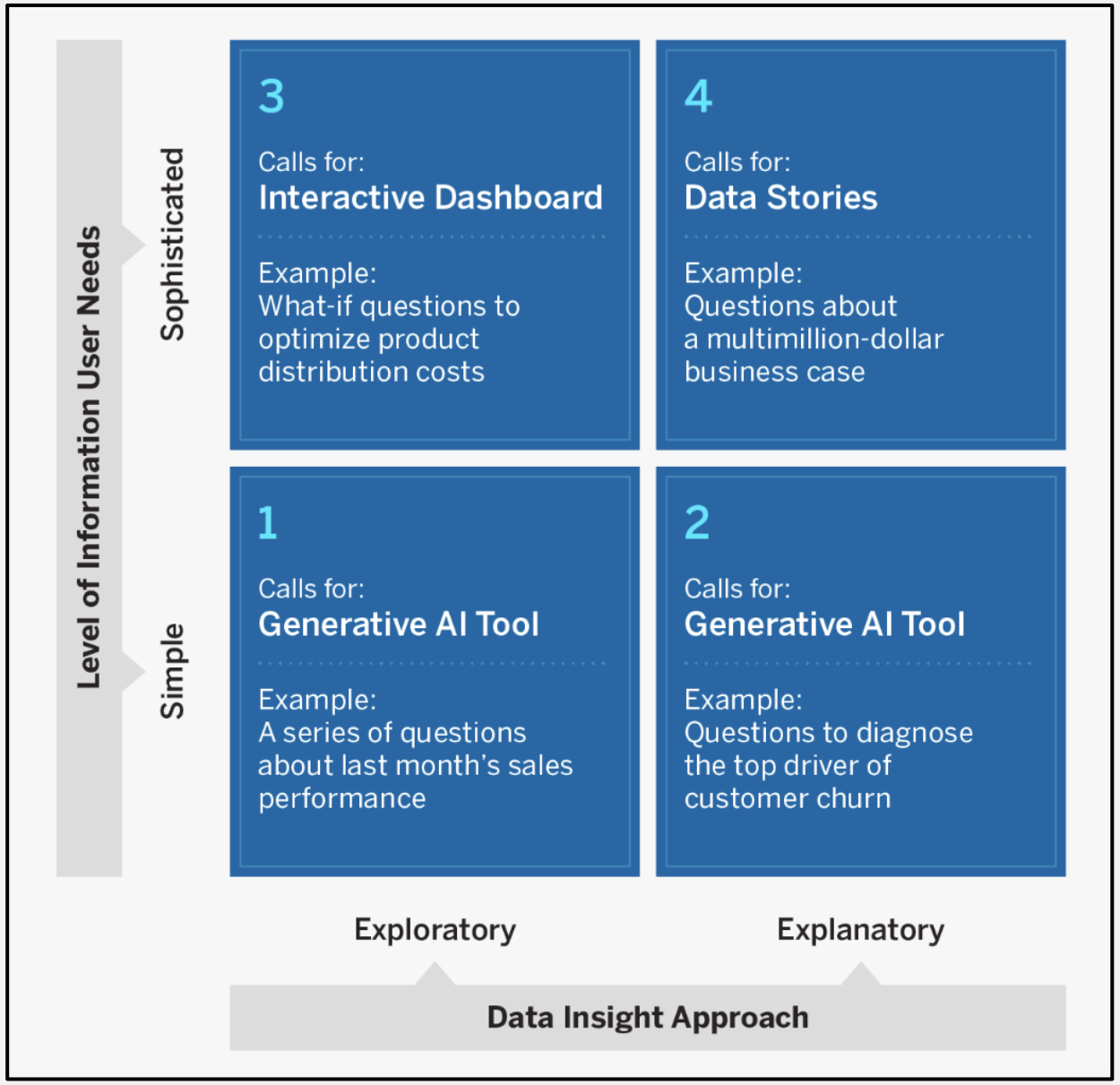
Did you do the pre-reading for today?

MIT Sloan Management Review

The Enduring Power of Data Storytelling in the Generative AI Era

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The Enduring Power of Data Storytelling in the Generative AI Era



Check out voice-activated GenAI applied to data visualization in my demo here.

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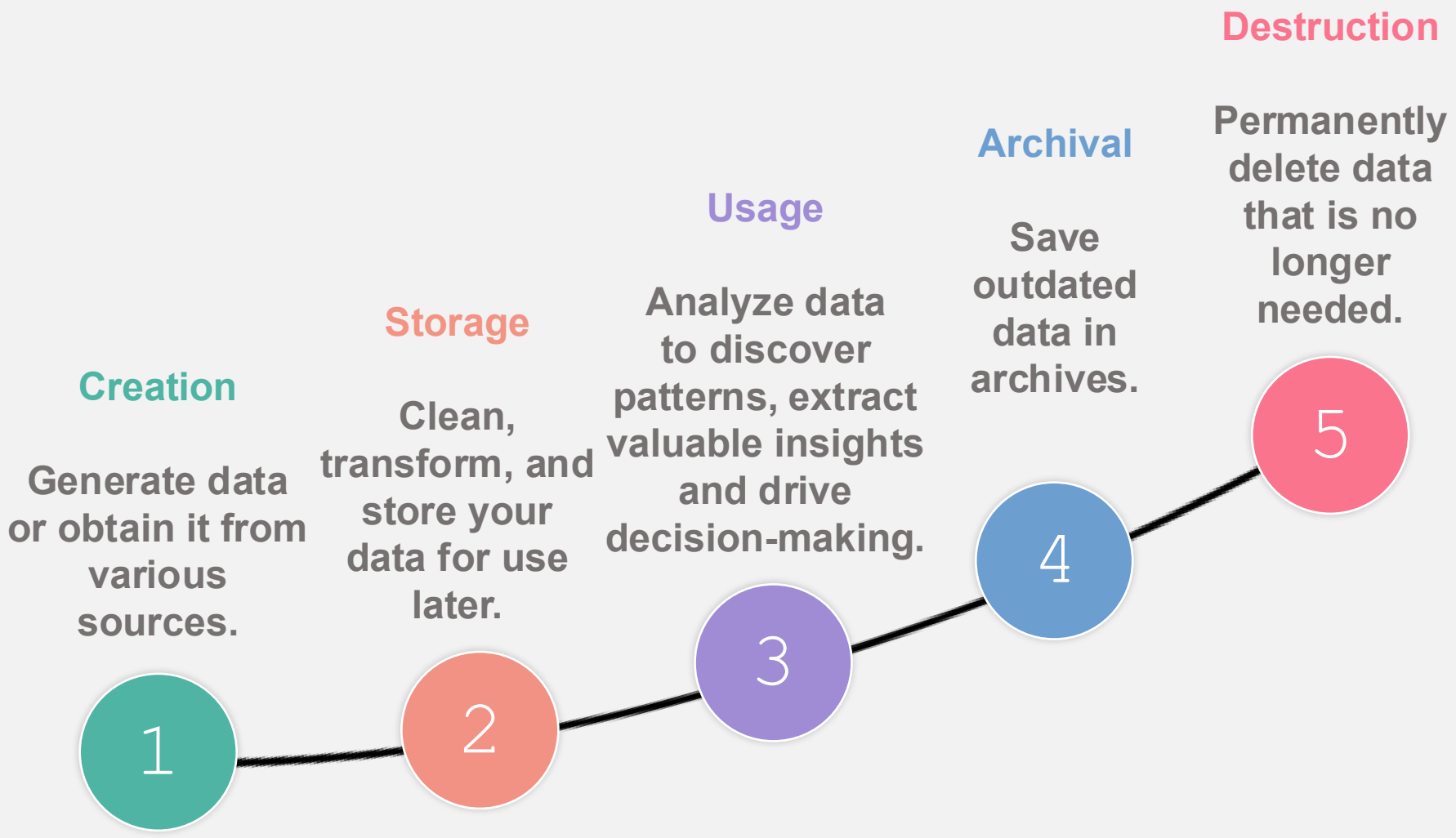
Data Lifecycle

From creation to destruction



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The Data Lifecycle



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1 CREATION

Important to consider: Quality & Relevance!

Filtering out unusable data will help you create a more manageable dataset that is cheaper to store.

Generate data or obtain it from various sources.

- Databases
- Web Analytics
- Data Entry Forms
- POS Sales Data
- Cloud Data
- Apps
- Surveys
- IoT Sensors
- 3rd party Vendors
- Social Media

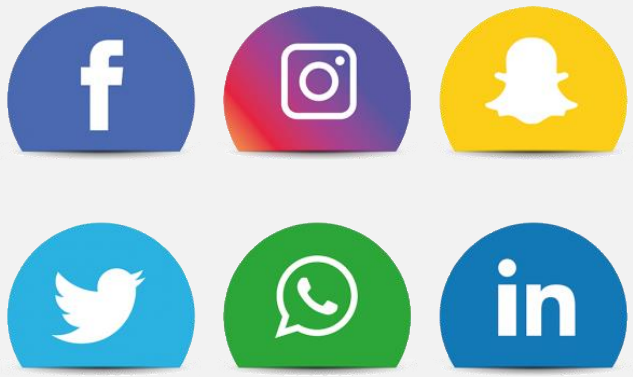
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1 CREATION

Structured vs. **Unstructured**

Varsity Sport	Males	Females	Total Participants
Baseball	29	0	29
Basketball	14	14	28
Cross Country	16	18	34
Lacrosse	35	19	54
Soccer	29	24	53
Swimming	29	33	62
Tennis	10	10	20
Track and Field	34	23	57
Wrestling	37	0	37
Softball	0	16	16
Volleyball	0	16	16













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2

STORAGE (& PROCESSING)

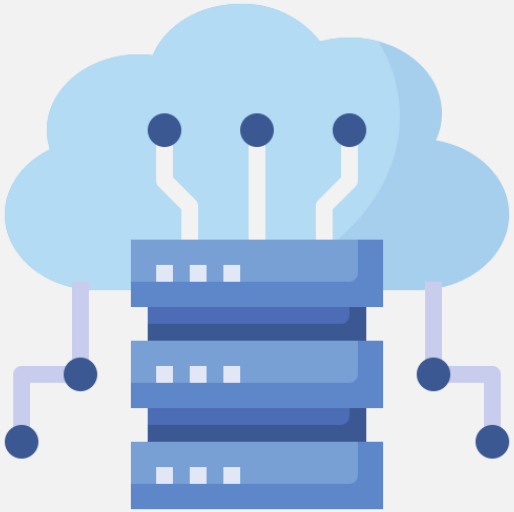
Clean, transform, and store your data for use later.

Manual	On-Premise	Cloud









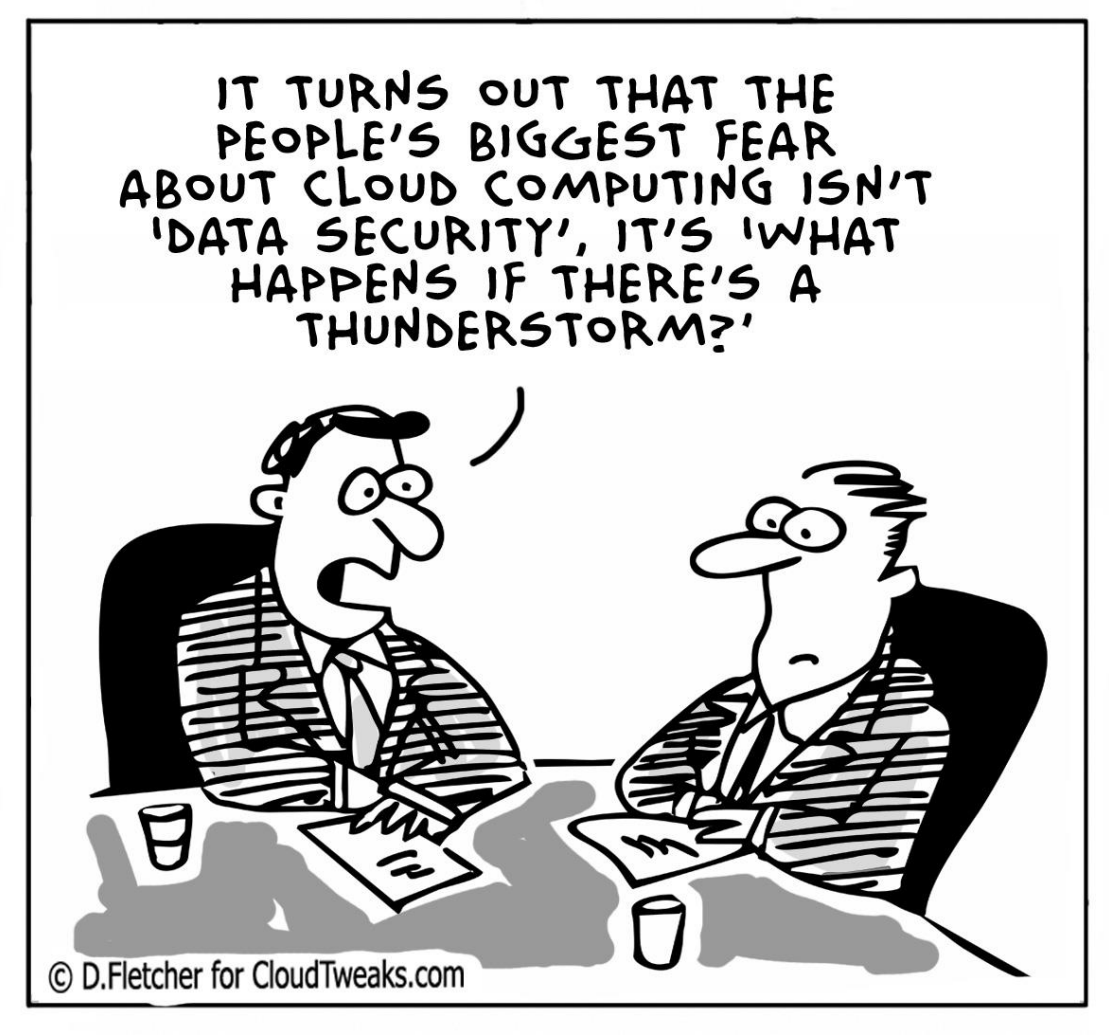
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2 STORAGE

The Cloud





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2 STORAGE

Where is the Internet hosted? Google Data Centers

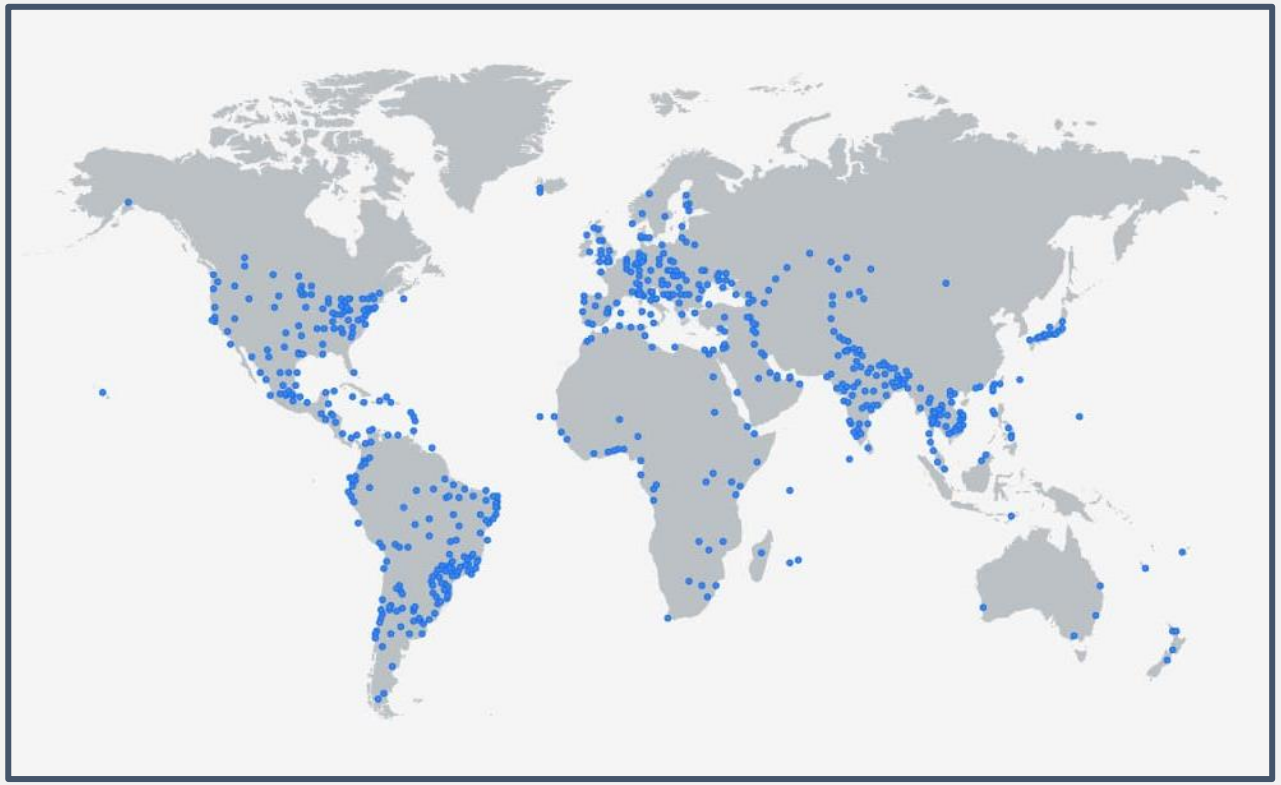


Image by Google

Google has over 130 data centers around the world.

(rapidly expanding thanks to GenAI hype)

...

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2 STORAGE

Where is the Internet hosted? Google Data Centers



Image by Google

Google has over 130 data centers around the world.

(rapidly expanding thanks to GenAI hype)

Hamina, Finland is one of them, located in an old paper factory & uses 100% renewable energy. Water from the ocean is used to cool the servers.

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2 STORAGE

…and Microsoft?



Image by Microsoft

Microsoft has 300+ locations around the world, spanning across 30+ countries (linked by over 175,000 miles of fiber optic cables!).

In 2018, they submerged a data center to process workloads on the seafloor near Scotland, it saves on cooling and uses 100% renewable energy.

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What about Amazon?



Amazon has 100+ locations around the world that cover 245 countries and territories.

Northern Virginia, US is home to the largest data center hub at AWS.

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2 STORAGE

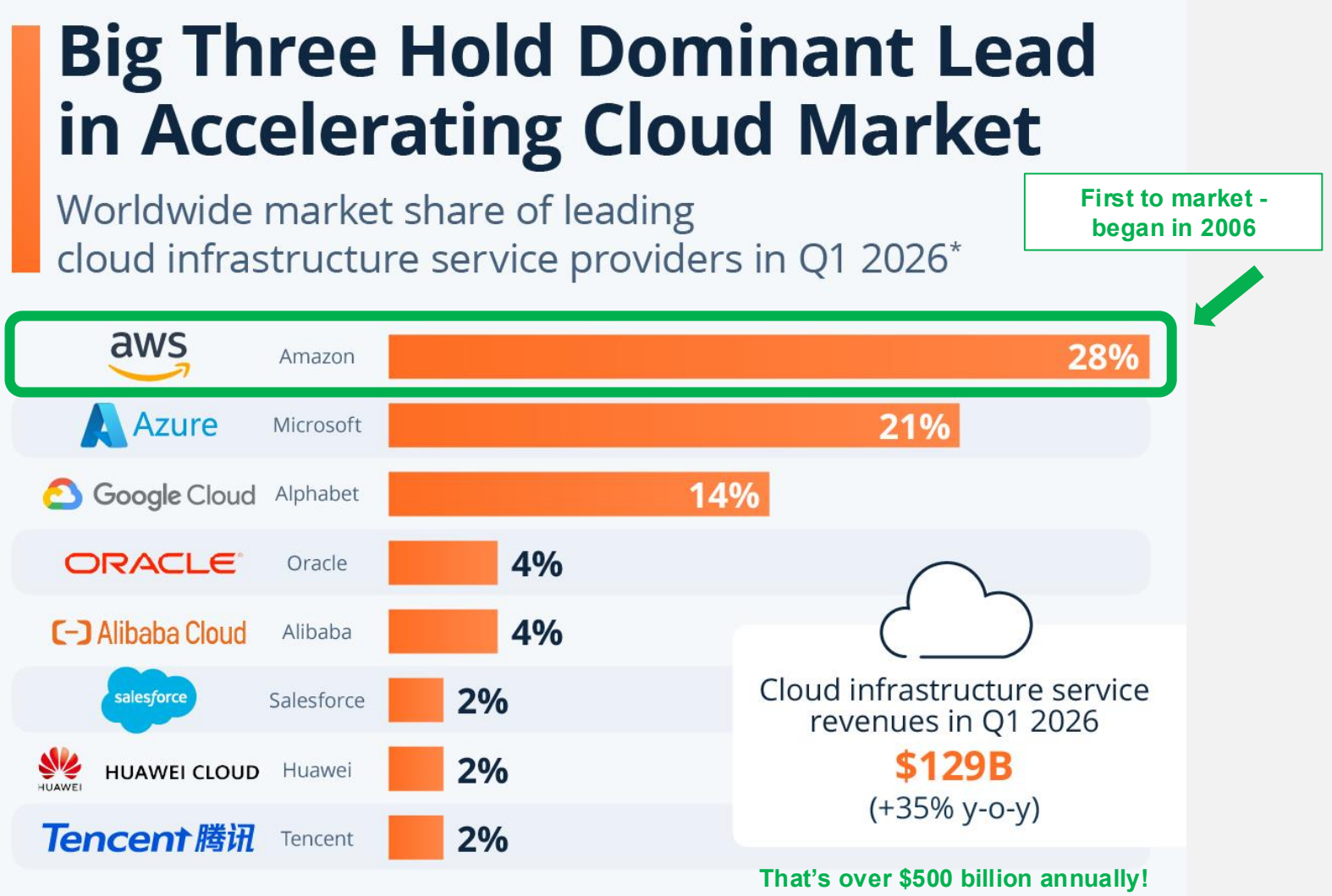
Top Cloud Providers Q3'2025

The Big Three

aws

Azure

Google Cloud



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2 STORAGE

Is Cloud always the best option for a company?

Nope!

- Does NOT always save on costs
  e.g. when working with predictable workloads that do not fluctuate in demand and require always consistent resources

- Where exactly data is stored can become a problem
  e.g. geopolitical events or changing regulations can restrict where data can be stored, especially across borders (China does not allow their citizen data to be stored outside of Chinese territory).

- Vendor lock-in is a serious risk
  e.g. dependence on a single cloud provider can limit flexibility and increase switching costs in the future

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2 STORAGE

Database

Typically referring to a relational database.

Designed to capture and record transactional data (OLTP - Online Transaction Processing).

Data stored in tables (rows and columns) – structured data!

Highly detailed.



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2 STORAGE

Data Warehouse

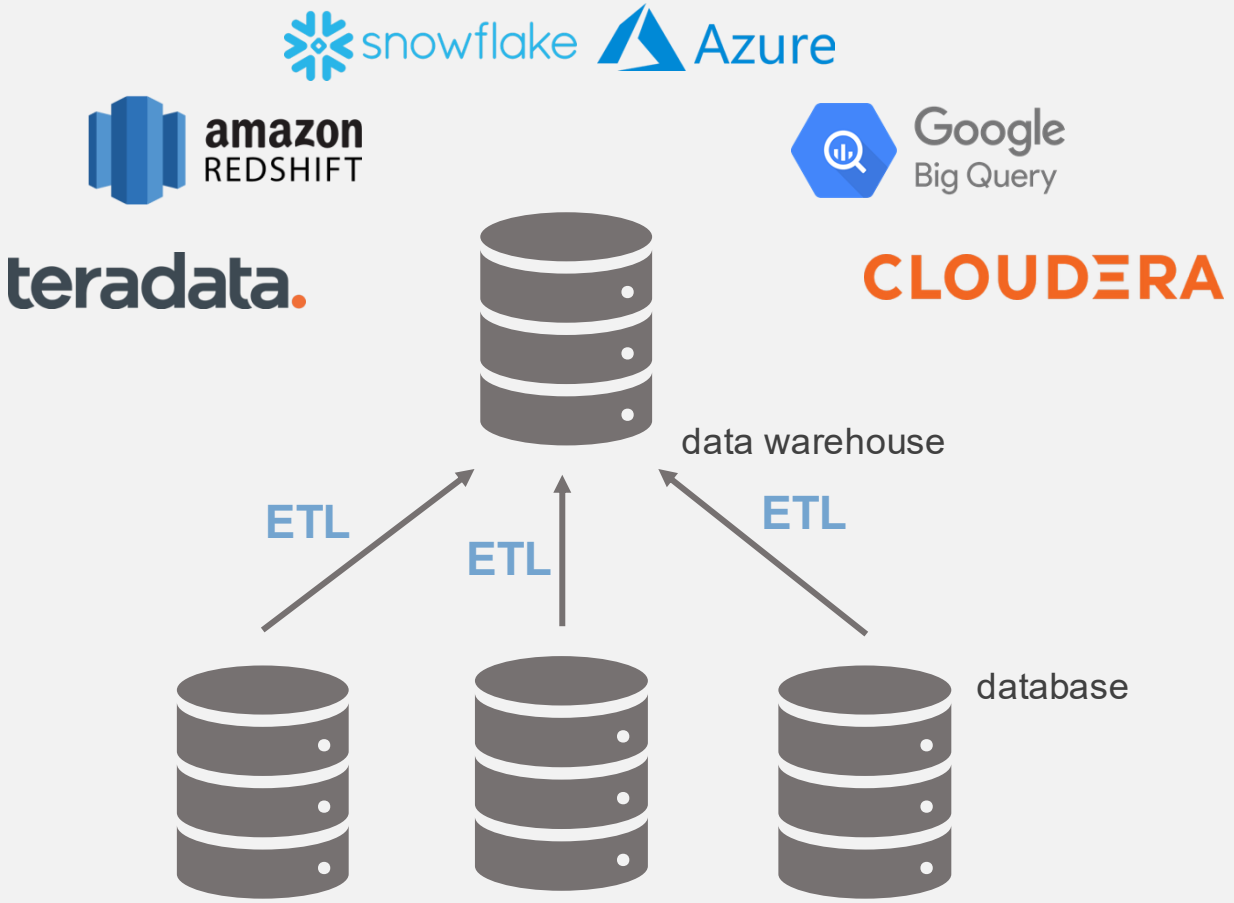
Typically referring to a relational database.

Designed for online analytical processing (OLAP).

Data is refreshed from database source systems.

Summarized data depending on business needs.

Business intelligence solutions.



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2 STORAGE

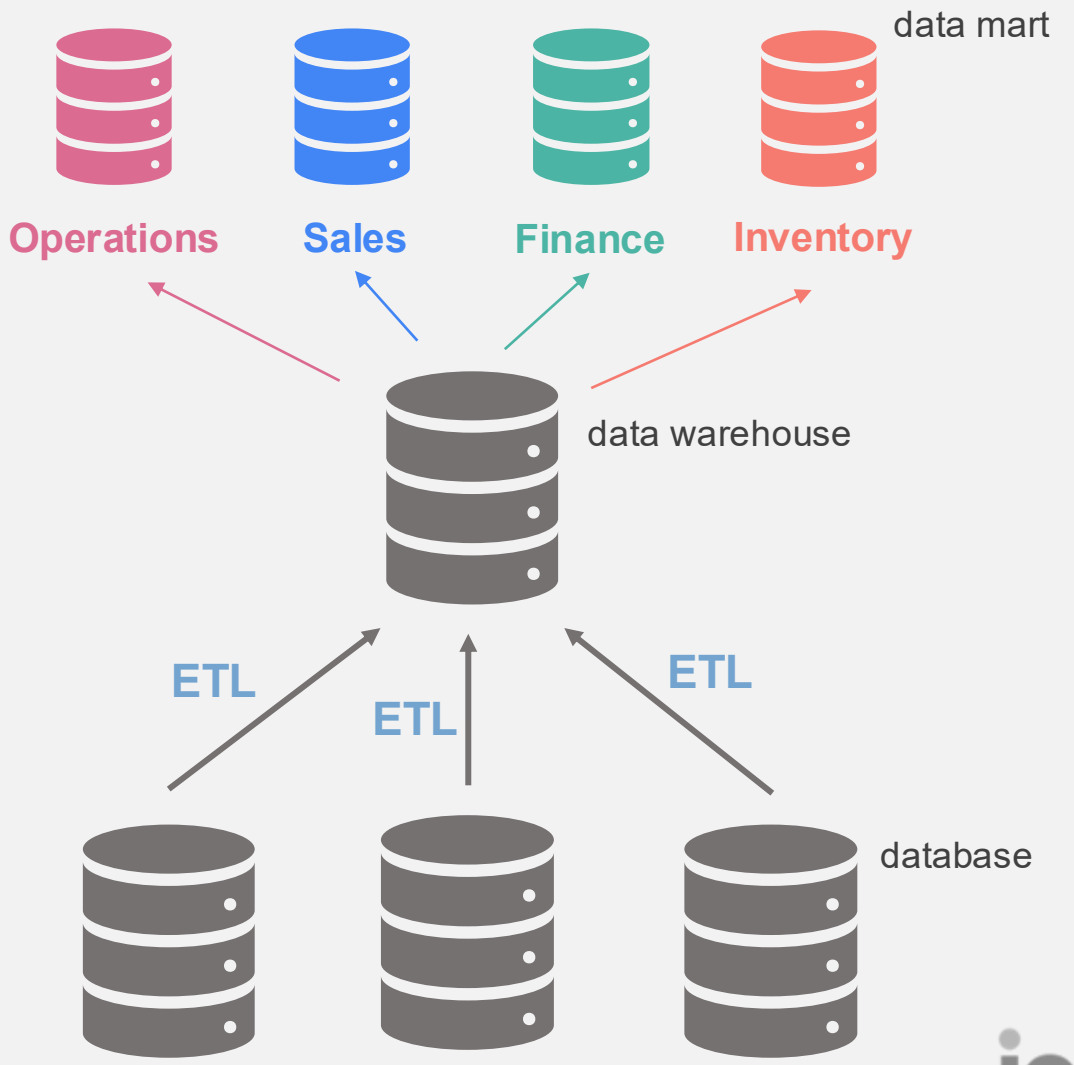
Data Mart

Subset of a data warehouse focused on a single line of business, department, team, or subject area.

Designed for use by that business line.

Small in size and more flexible than a large data warehouse.

Allows for quick and easy access data querying.



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2 STORAGE

Data Lake

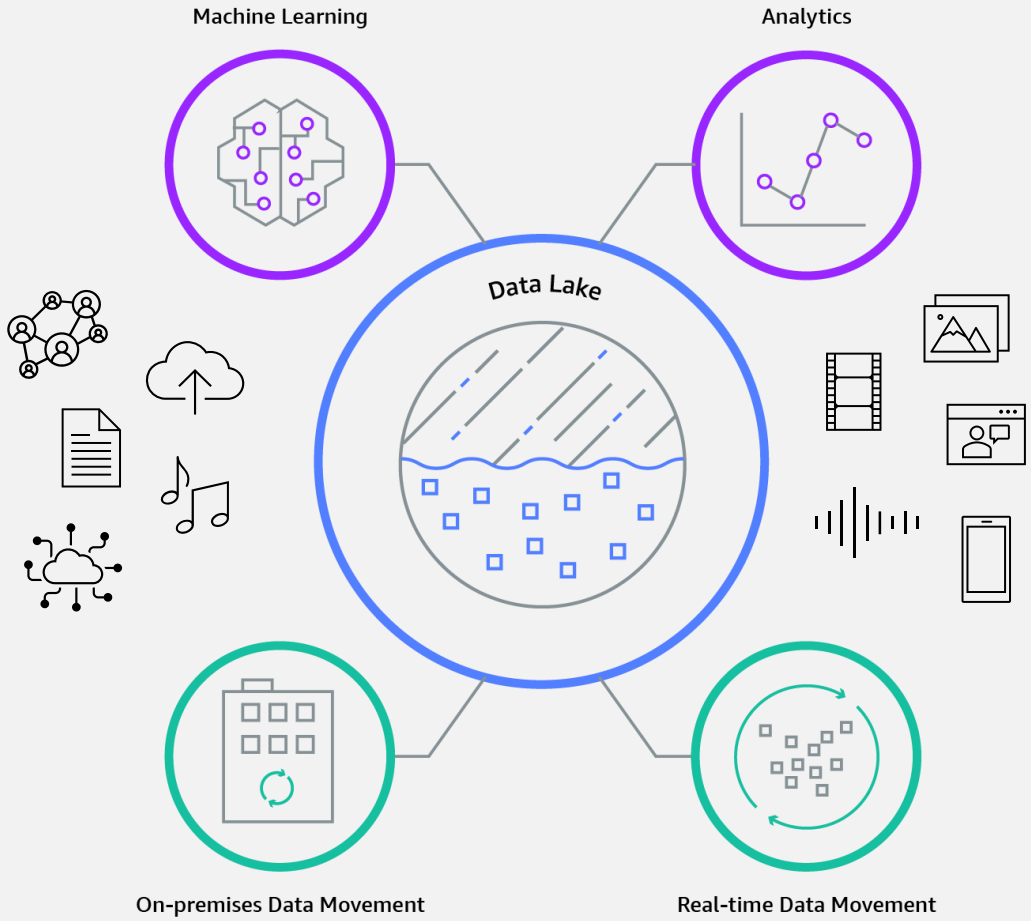
Designed to capture any raw data (structured, unstructured).

Made for large amounts of data.

Great for ML and AI using raw data or analytics if you process the data first.

Can be organized into databases or data warehouses (for the structured data).

Typical AI solution.



Microsoft Azure Data Lake

HPE GreenLake

Google Cloud

amazon web services S3

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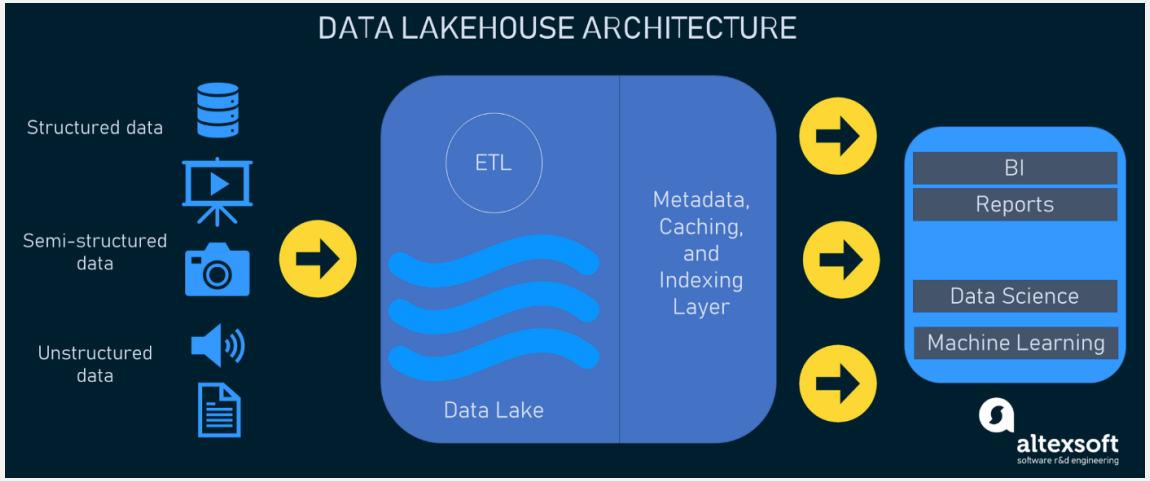
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2 STORAGE

Data Lakehouse

Convergence of a data warehouse and data lake (the best of both worlds!).

System design that combines transaction process of a data warehouse with the low-cost storage of a data lake.



dremio

ORACLE CLOUD

snowflake

databricks

ahana

Google Cloud BigLake

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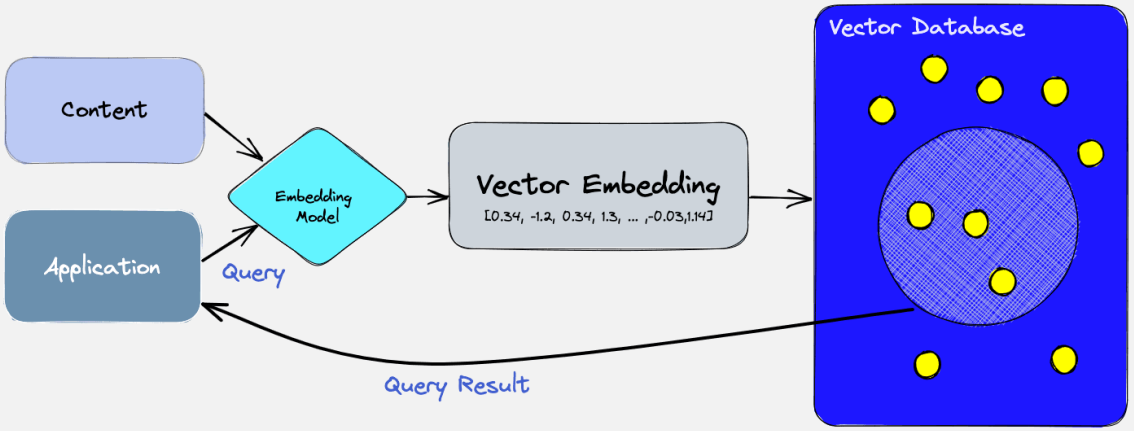
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2 STORAGE

Vector Database

Stores vectorized data – numerical representations of complex data like text, images, audio, video, etc.

Vector embeddings capture the relationships and meaning of content so computers can understand and process them effectively.



SingleStore™ Vertex AI Pinecone

Chroma elasticsearch

mongoDB Atlas

■ = vector database

● = search area *(relevant results for a given query)*

● = individual vector embeddings

SIMPLISTIC EXAMPLE
cat = [1.8, -0.5, 7.3, 21.7, 16.5]
dog = [1.5, -0.3, 6.9, 19.2, 17.2]

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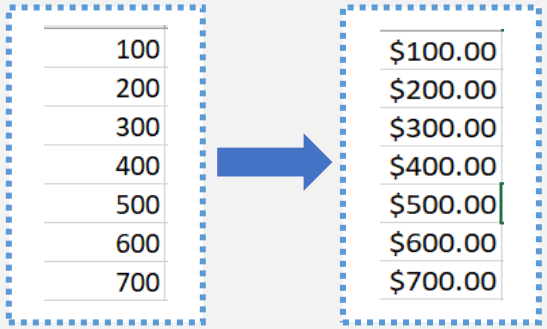
2

STORAGE (& PROCESSING)

We must process & cleanse data to ensure it's structured in a useful and business-ready way

Formatting

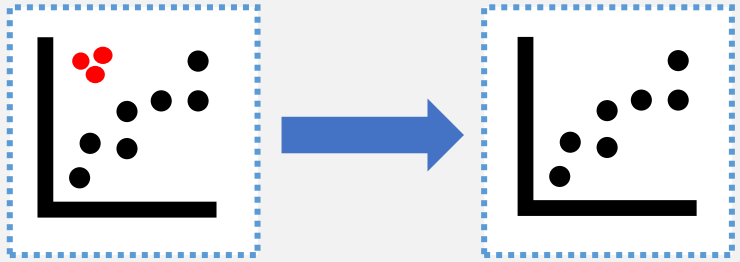
Convert data into a consistent and structured format



Example: Formatting financial data into consistent currency formats

Cleansing

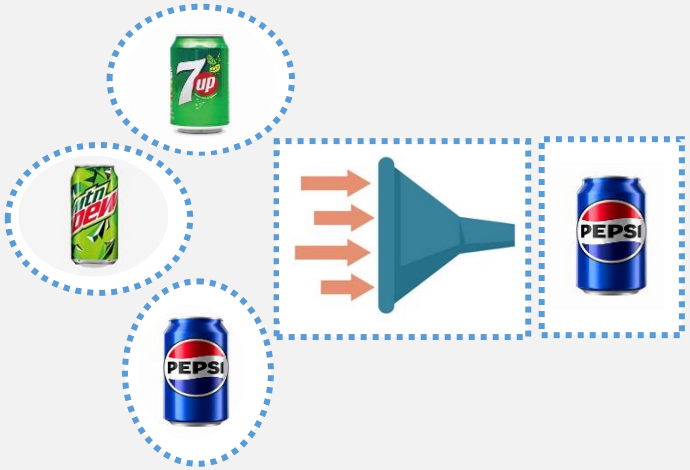
Identify and correct errors or inconsistencies in the data



Example: Addressing outliers in data that may skew the analysis

Filtering

Select a subset of data based on specific criteria



Example: Refining filters based on business needs for an analysis

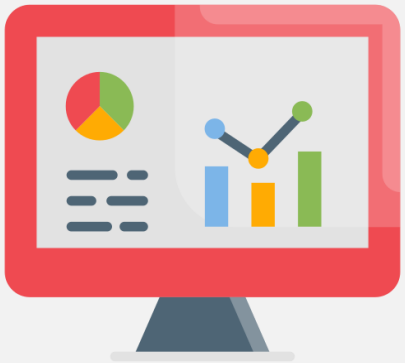
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3 USAGE

We will be focusing most of our course on this point.

Analyze data to discover patterns, extract valuable insights, and drive decision-making.



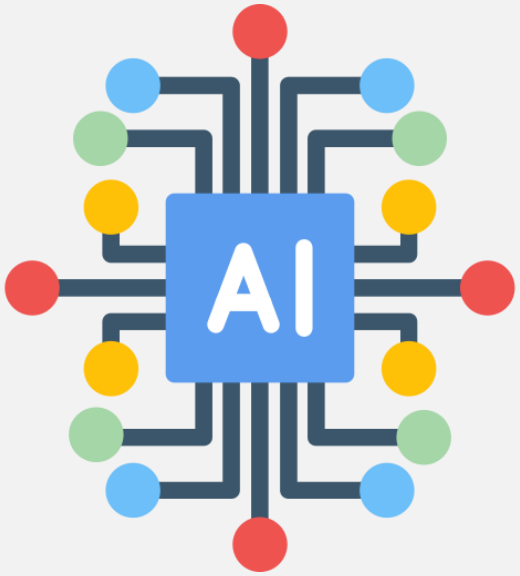
Dashboards & Reporting



Market Research



Machine Learning



Artificial Intelligence

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4 ARCHIVAL

Once data has fulfilled its main purpose and become outdated, it may still need need to be saved for legal, regulatory or historical reasons.



Often done with archiving software, which moves aging data to archives according to the corporate archival policy.

FROM		TO
Production Systems	→	Long-term Storage

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5 DESTRUCTION

Archived data that is no longer needed is permanently deleted to lower risk, costs, and obligations of unnecessary data storage while freeing up space.

Regulations cover data disposal in some industries, like healthcare, and can result in hefty fines if not followed closely.

Well-documented data destruction procedures should be drafted at an organizational level.



FROM → TO → TO

Production Systems → Long-term Storage → Destruction

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Feel free to stay in touch!

(I've reached the 30K connection limit on LinkedIn unfortunately, so until I can get that back down, you can follow me there!)

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